Basics of Windows server and AD theory

```
Hypervisor          : Oracle VirtualBox
VM name             : Server01
Operating system    : Windows Server 2019 Datacenter Evaluation
Computer name       : Lab01  (FQDN: lab01.lab.com)
IPv4 address        : 10.10.10.1   Subnet mask: 255.0.0.0
Domain (forest)     : lab.com
```

Task 1: Install Windows Server on a virtual machine using VirtualBox or VMware and take a screenshot of the desktop after the installation completes.

I created a new virtual machine named Server01 in Oracle VirtualBox, attached the Windows Server 2019 ISO and started the VM. Windows Setup then copied the files and installed Windows. I have attached two screenshots below showing the installation in progress.

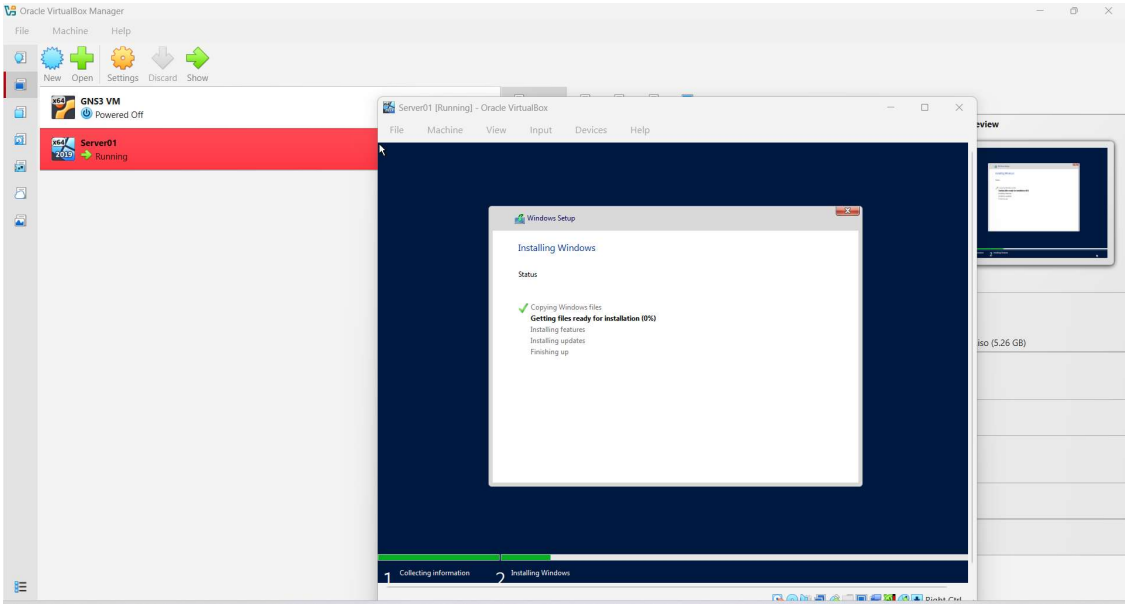


Figure 1: Windows Setup, getting files ready for installation (0%)

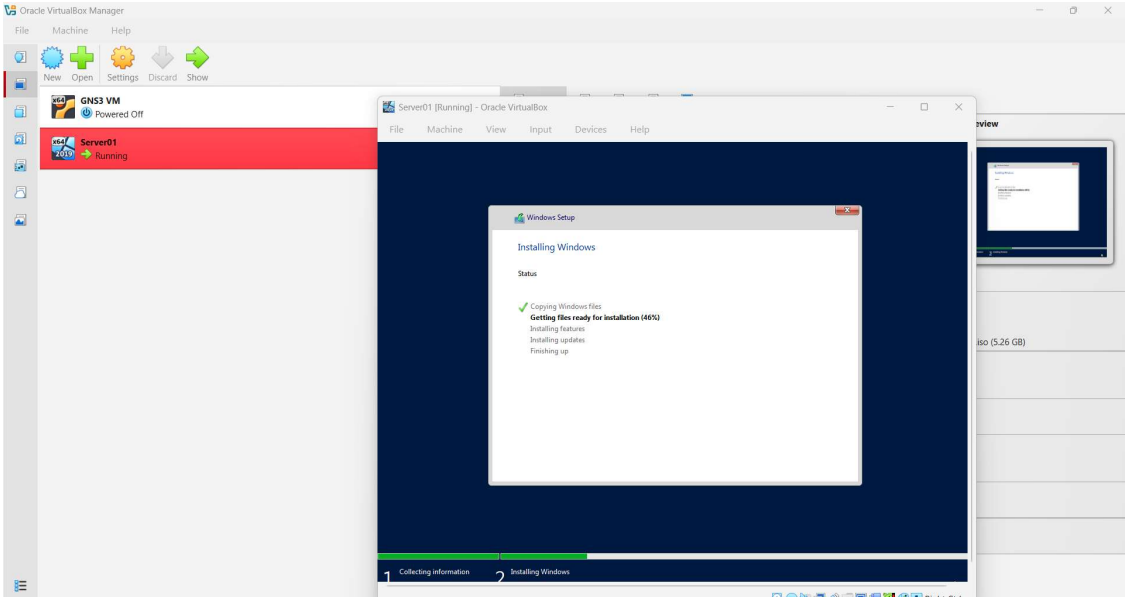


Figure 2: Windows Setup, getting files ready for installation (46%)

Task 2: Open Server Manager and use the Add roles and features wizard to install the Active Directory Domain Services (AD DS) role. Document each step with a short description and screenshot.

Step 1: Start the wizard

In Server Manager I clicked Manage > Add Roles and Features. The Before You Begin page reminds you that the administrator account needs a strong password, the network settings (such as a static IP) must be configured, and Windows updates should be installed. I clicked Next.

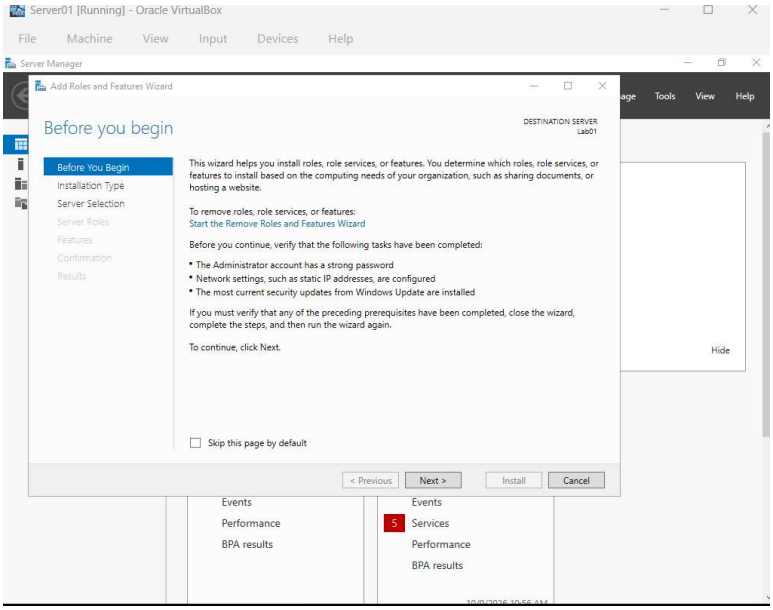
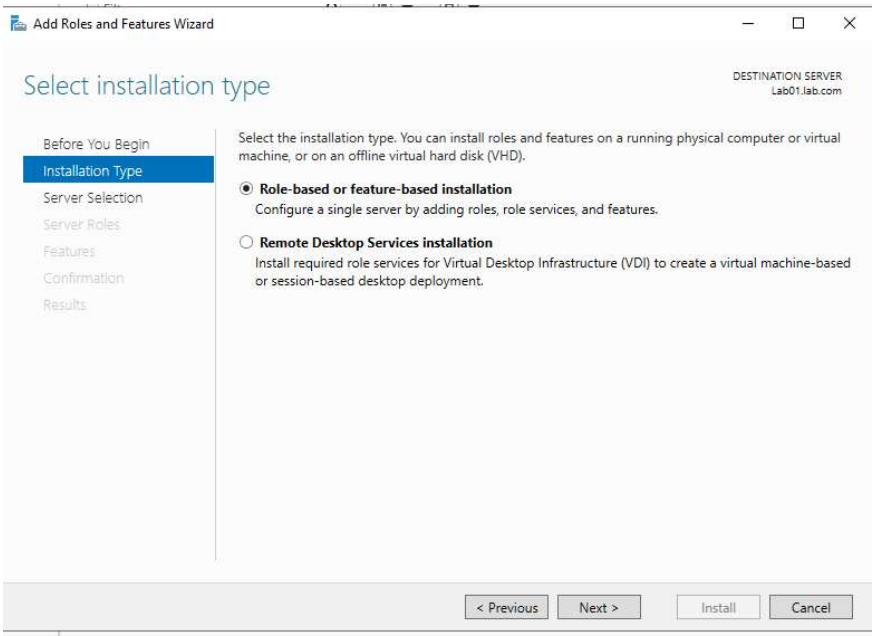


Figure 3: Add Roles and Features Wizard, Before You Begin

Step 2: Choose the installation type



Step 3: Select the destination server

I selected "Select a server from the server pool" and chose Lab01 (10.10.10.1, Windows Server 2019 Datacenter Evaluation).

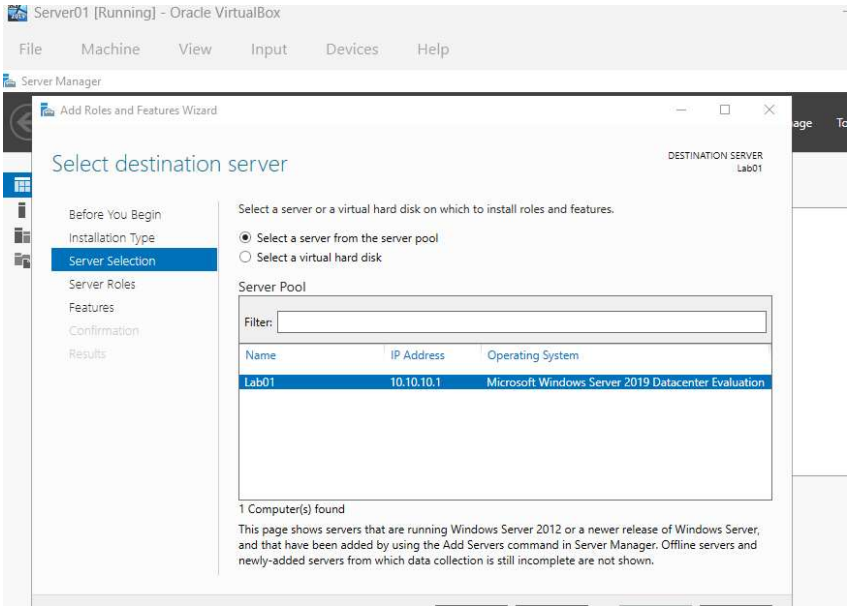


Figure 4: Select destination server

Step 4: Select the server role

On the Server Roles page I ticked Active Directory Domain Services. The description on the right explains that AD DS stores information about objects on the network and makes it available to users and administrators.

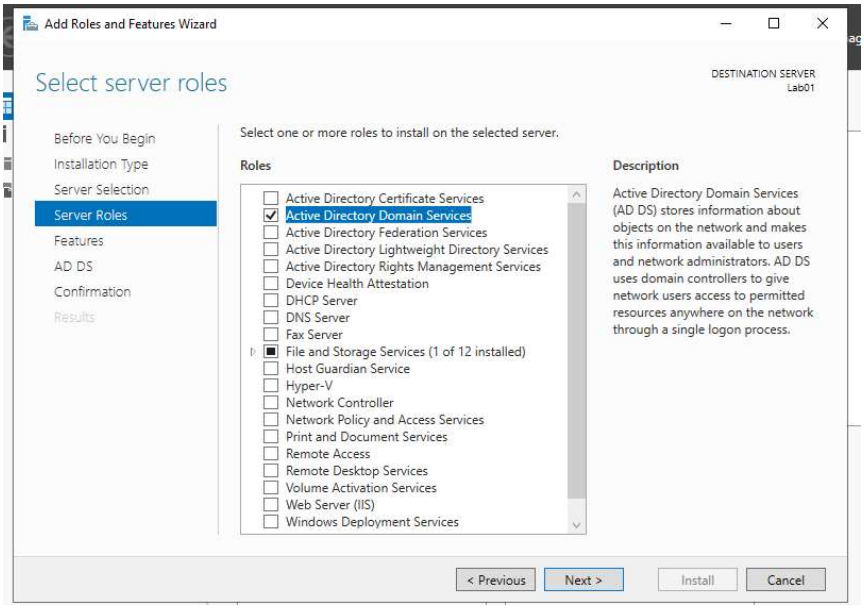


Figure 5: Select server roles, Active Directory Domain Services ticked

Task 3: Promote your Windows Server to a Domain Controller by creating a new forest with a root domain name of your choice (e.g., mycollege.local). List the options you selected during the configuration.

After the AD DS role finished installing, I used the "Promote this server to a domain controller" option in Server Manager to start the Active Directory Domain Services Configuration Wizard.

Options selected

- Deployment operation: Add a new forest
- Root domain name: lab.com

- DNS server: installed together with the domain controller (the DNS zone lab.com was created)
- Database folder and log files folder: C:\Windows\NTDS
- SYSVOL folder: C:\Windows\SYSVOL
- Forest and domain functional level: Windows Server 2016 (both forest and domain)
- Directory Services Restore Mode (DSRM) password: set (not shown in this report)

Paths page

I kept the default locations for the AD DS database, log files and SYSVOL.

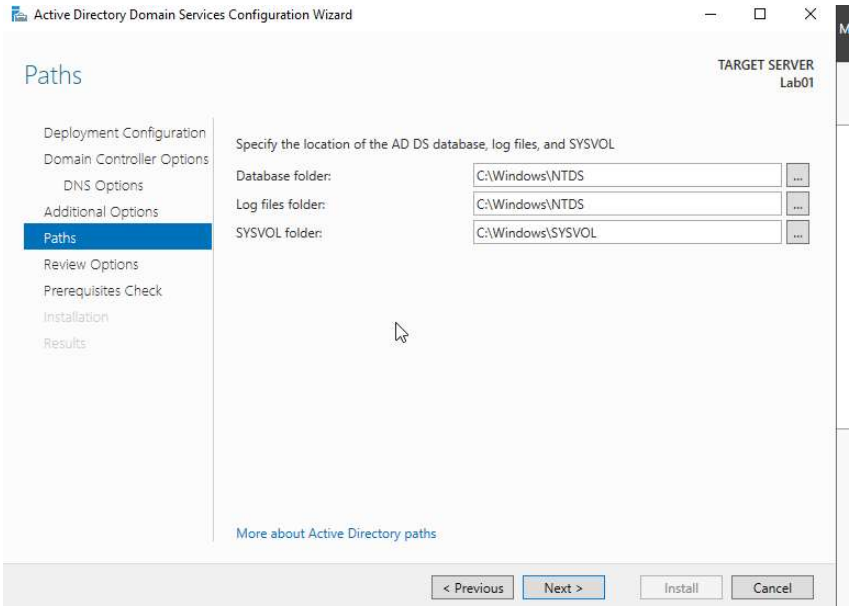


Figure 6: AD DS Configuration Wizard, Paths

Verification

After the restart, the server (Lab01) appears in Active Directory Administrative Center as a Global Catalog domain controller in the lab.com domain. DNS Manager shows the forward lookup zone lab.com, and nslookup resolves both lab01.lab.com and lab.com to 10.10.10.1.

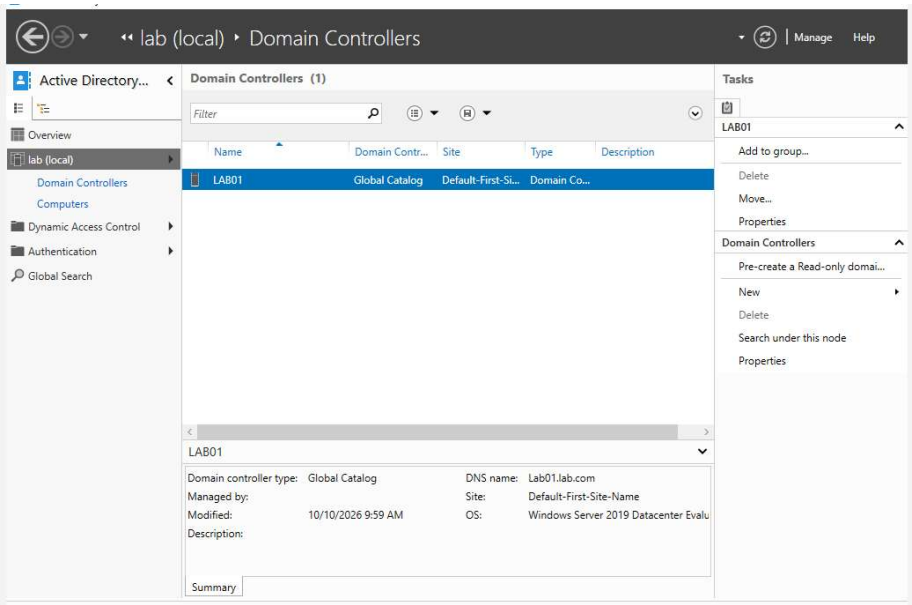


Figure 7: Domain Controllers container showing LAB01 (Global Catalog)

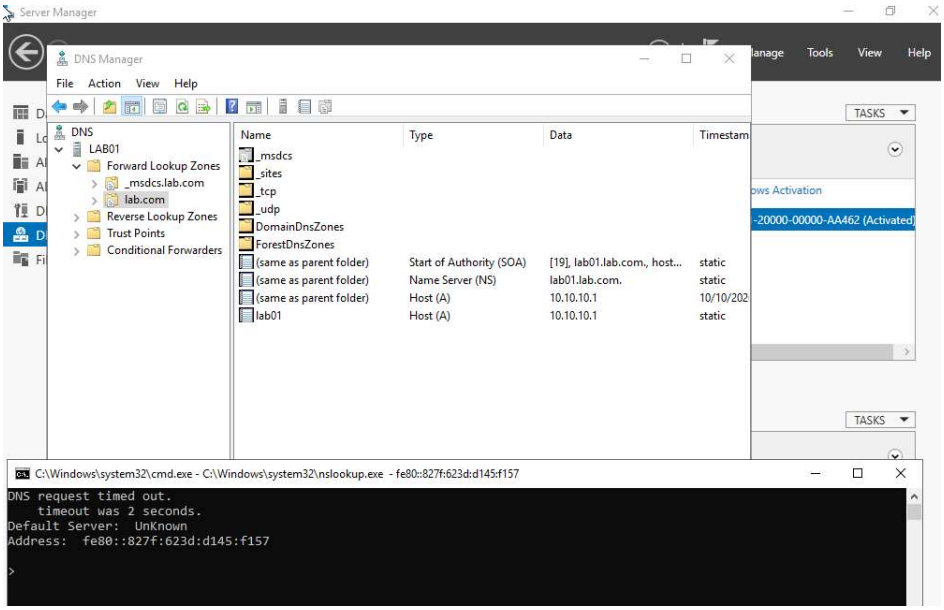


Figure 8: DNS Manager, forward lookup zone lab.com with the host record for lab01 (10.10.10.1)

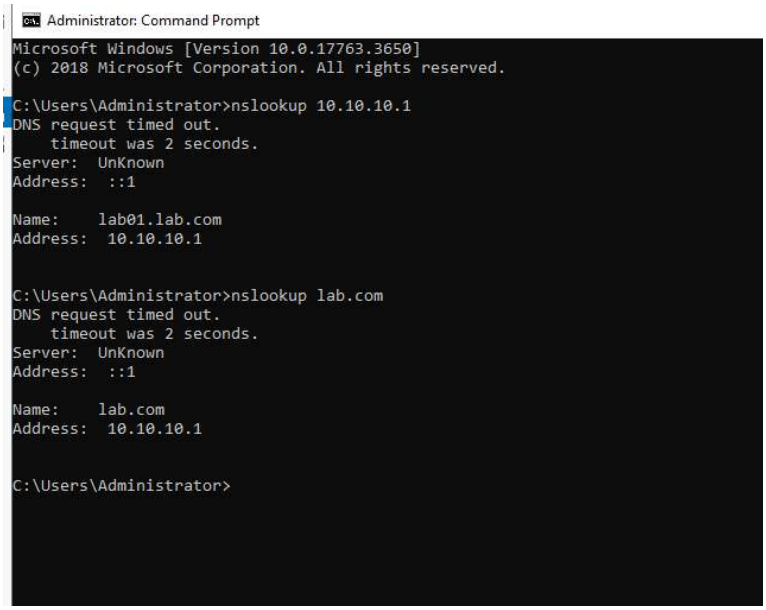


Figure 9: nslookup for 10.10.10.1 and lab.com from the server

Task 4: Create two new user accounts and one group in Active Directory Users and Computers. Assign both users to the group and describe how you did it.

I used Active Directory Administrative Center (ADAC), which manages the same directory objects as Active Directory Users and Computers (ADUC). Both tools can be used for this task.

Step 1: Create an organizational unit

In ADAC I right-clicked the domain lab (local) and chose New > Organizational Unit. I entered the name HOD Users, left "Protect from accidental deletion" ticked, and created it in DC=lab,DC=com.

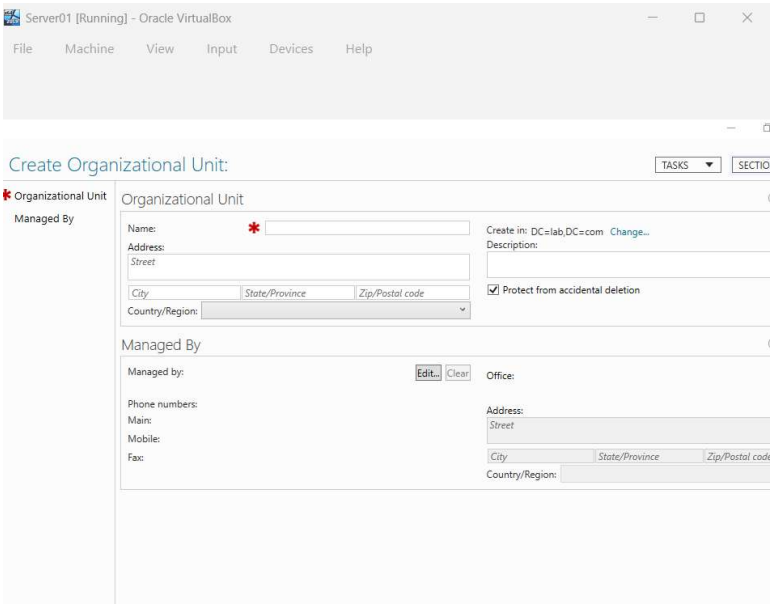


Figure 10: Create Organizational Unit form, blank

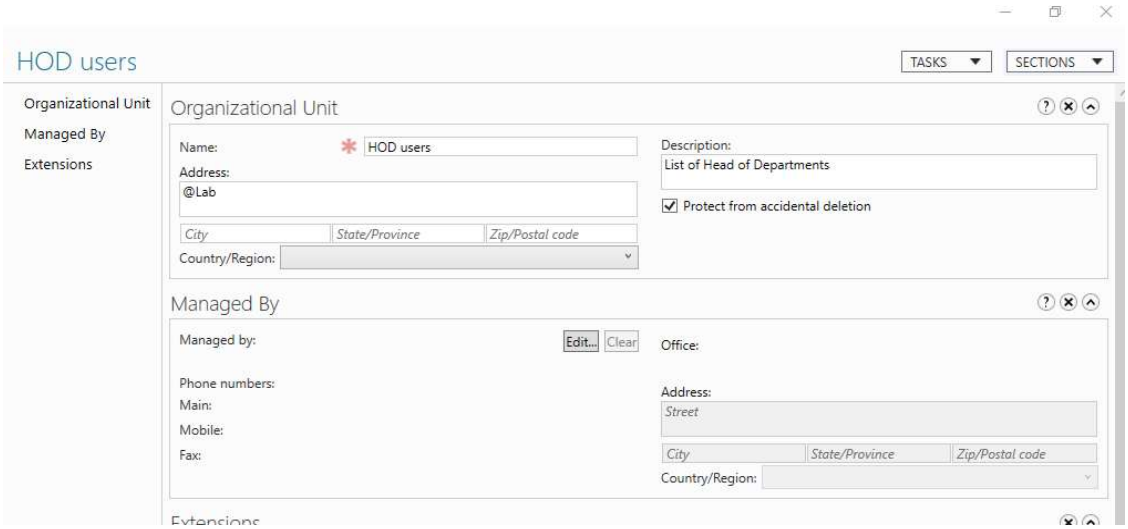


Figure 11: Create Organizational Unit: HR (Create in: DC=lab,DC=com)

Step 2: Create the two users

Inside the HR OU I chose New > User and created two accounts. For each one I filled in the name, the user logon name (UPN lab.com) and the password, and set "Password never expires". I also filled in the Organization section (job title and department).

```
User 1: John Doe   logon: john.d@lab.com   Job title: Head of Department   Department: HR
User 2: Mira M     logon: mira.m@lab.com   Job title: HOD                  Department:
Accounts
```

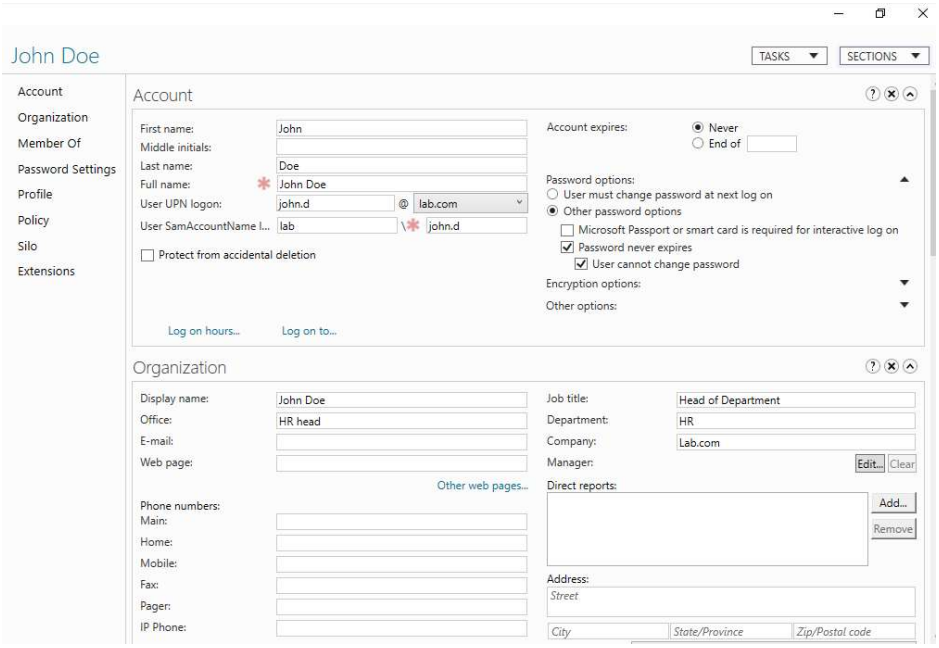


Figure 12: User account John Doe

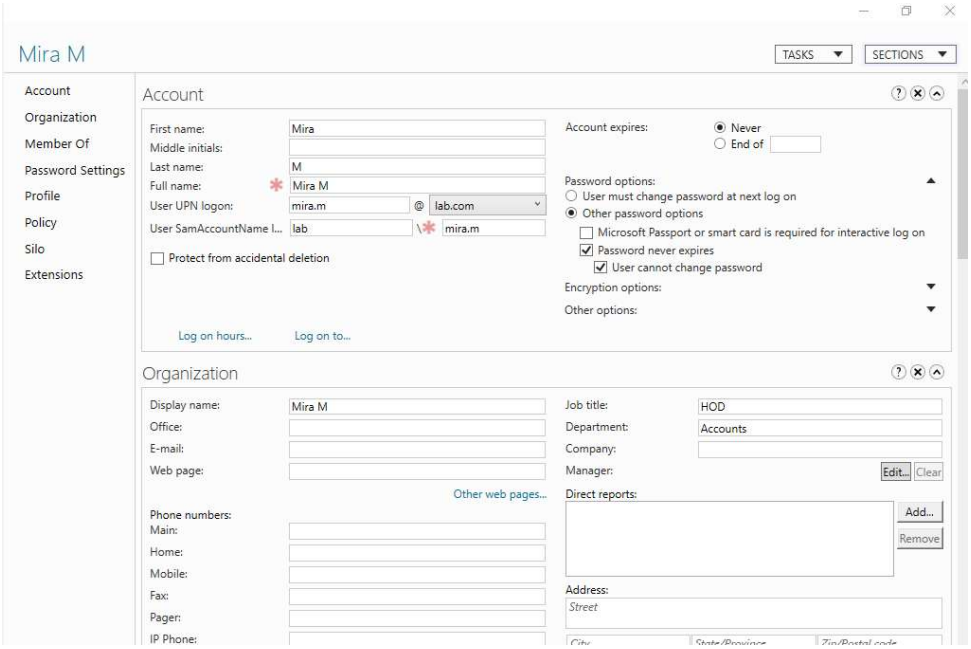


Figure 13: User account Mira M

Step 3: Create the group

In the HR OU I chose New > Group and named it HOD. I set the group type to Security and the scope to Global. A Security group is used so the group can be given permissions to resources, which a Distribution group cannot do.

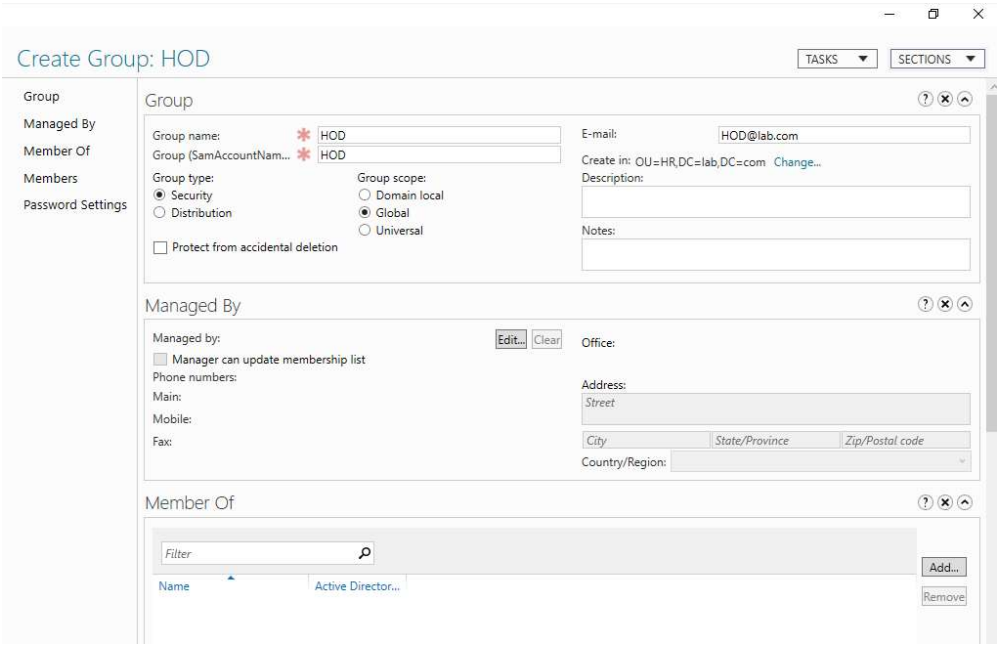


Figure 14: Create Group: HOD (Security, Global, created in OU=HR)

Step 4: Add both users to the group

In the Members section of the group I clicked Add and selected John Doe and Mira M. Both users now appear in the member list.

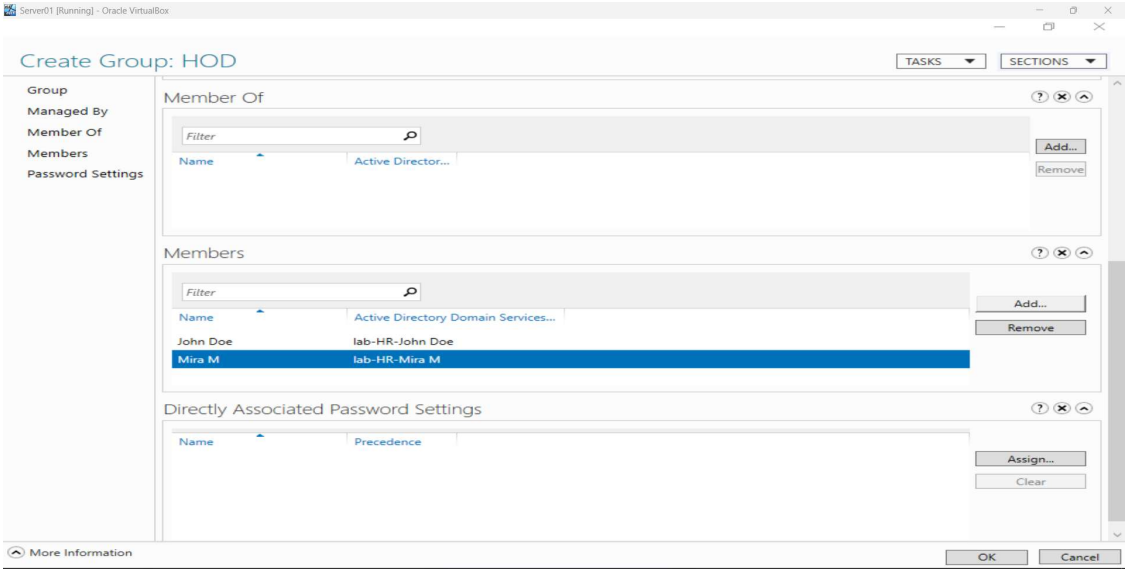


Figure 15: HOD group, Members section with John Doe and Mira M

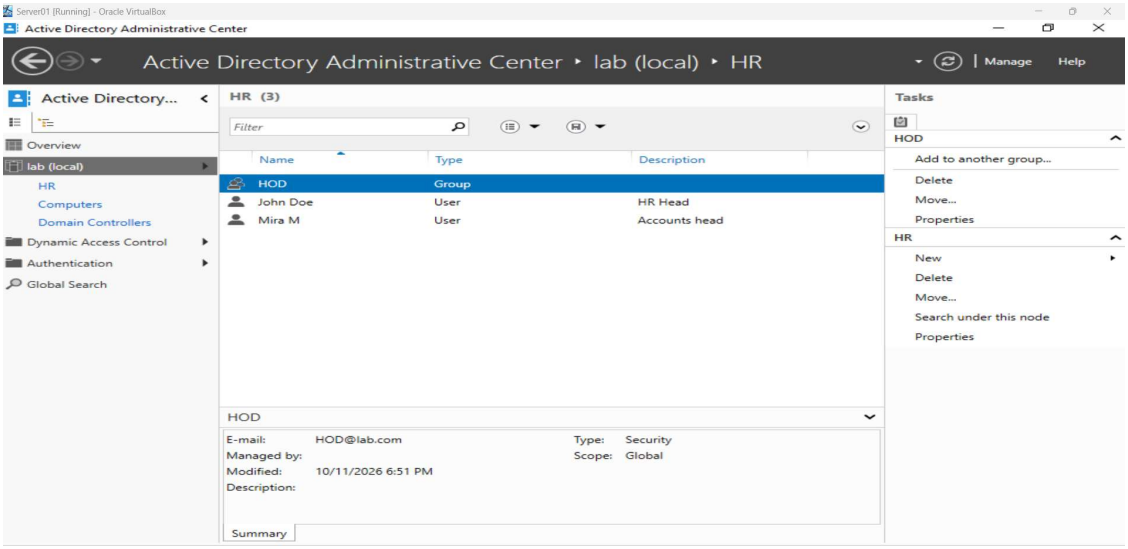


Figure 16: HR OU containing the HOD group and both users

Step 5: Reorganize into separate OUs

To keep the directory tidy I then created two more OUs: "HOD users" for the user accounts and "Group lists" for groups, and moved the objects into them using Move. The HOD group is now in "Group lists".

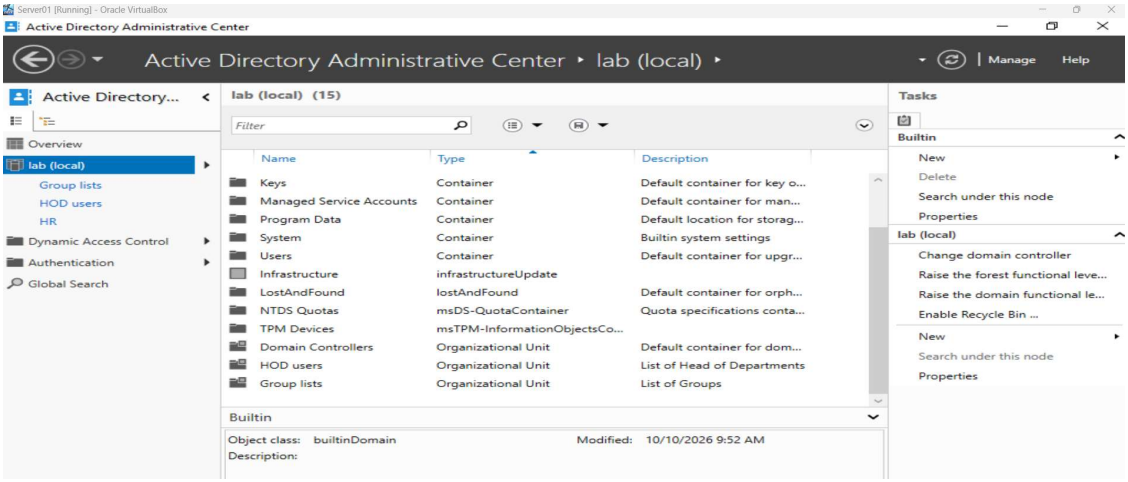


Figure 17: Domain root showing the OUs HOD users, Group lists and HR

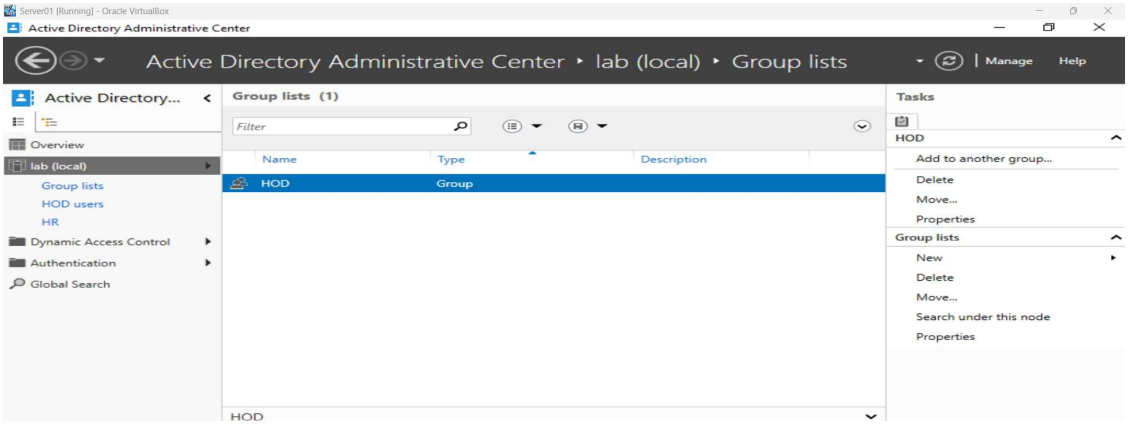


Figure 18: Group lists OU containing the HOD group

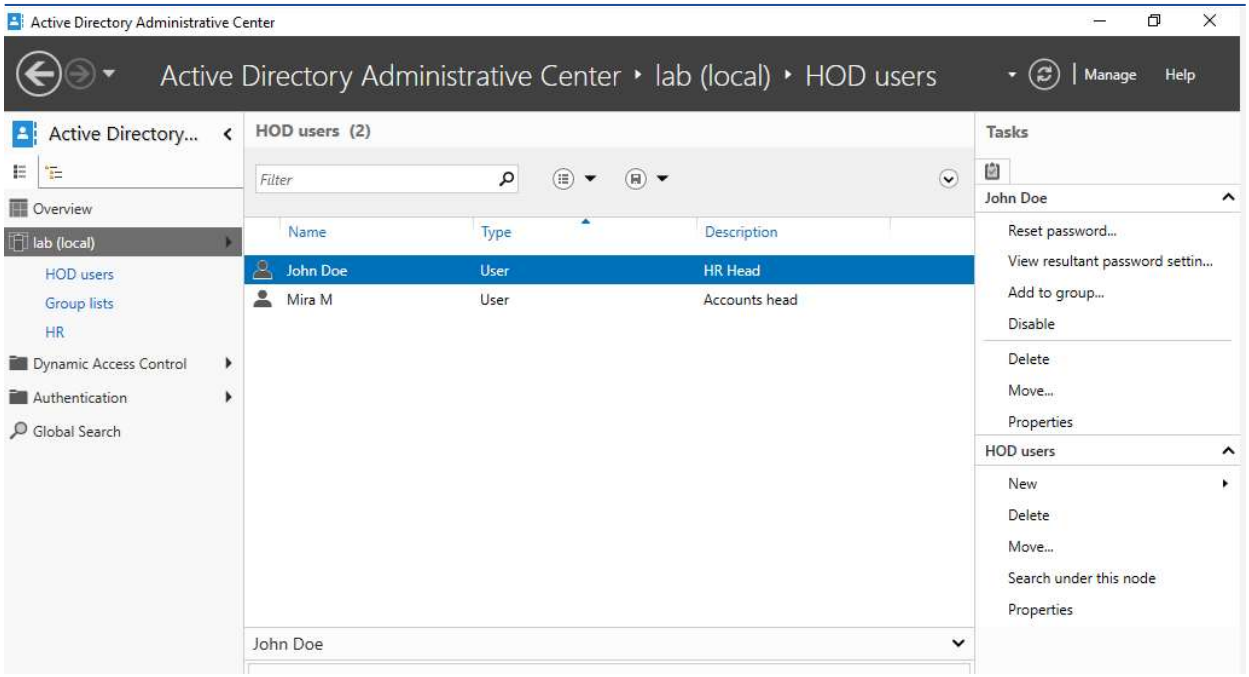


Figure 19: OU containing the HOD users list